Procédure Pfsense

Introduction – Objectif de la procédure réalisée

Dans le cadre de notre infrastructure virtualisée sous Proxmox VE, pfSense a été déployé en machine virtuelle afin de servir de routeur et pare-feu principal pour l'ensemble du réseau.
Une fois la configuration de base terminée (assignation des interfaces, obtention de l'adresse IP publique sur WAN, réseau LAN opérationnel), il a été nécessaire de rendre accessibles depuis l'extérieur (Internet) deux services critiques :
L'interface d'administration web de pfSense
L'interface web de Proxmox (port 8006)
Cependant, il est formellement déconseillé d'exposer directement les ports par défaut (443 et 8006) sur Internet en raison des risques élevés d'attaques
La procédure qui suit détaille donc la méthode mise en œuvre pour ouvrir ces accès de façon temporaire, mais en respectant les bonnes pratiques de sécurité minimales :

Etape 0 changer le mot de passe du compte admin

```
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

 WAN (wan) -> hn0 -> v4/DHCP4: 10.211.225.246/24
 LAN (lan) -> hn1 -> v4: 192.168.10.254/24

 0) Logout / Disconnect SSH           9) pfTop
 1) Assign Interfaces                10) Filter Logs
 2) Set interface(s) IP address      11) Restart GUI
 3) Reset admin account and password 12) PHP shell + pfSense tools
 4) Reset to factory defaults        13) Update from console
 5) Reboot system                    14) Enable Secure Shell (sshd)
 6) Halt system                      15) Restore recent configuration
 7) Ping host                        16) Restart PHP-FPM
 8) Shell

Enter an option: 3


The authentication configuration and privileges for the "admin" account will be reset
 to the default.
Proceed? (y|n): y

The default administrator account in the User Manager ("admin") has been reset.
The password must now be set to a new value.
Changing password for 'admin' in the User Manager database.
Hints:
* Current NIST guidelines prioritize password length over complexity.
* The password cannot be identical to the username.
New Password:
Confirm New Password:
```

Procedure Pfsens

Etape 1: associer WAN ET LAN a leur port respectif et création de VLAN 10

1. Sur CLI de pfsense en rentre commande :1
2. Should VLANs be set up no(y:n) : tape y
3. Warraning → tape (y) si ya pas de vlans deja configure
4. Do you want to proceed (y:n) → y
5. Parent interface → em1 (le port du LAN)
6. VLAN tag → dans mon cas j’ai met 10
7. Entre the WAN interface name → donne le nom du port qui lui correspond dans notre cas c’est em0 et pour le LAN em1
8. Enter the optional 1 interface name → la on met interface de LAN.10 (10 c’est le tag du VILAN) → enter
9.

```
If the names of the interfaces are not known, auto-detection can
be used instead. To use auto-detection, please disconnect all
interfaces before pressing 'a' to begin the process.

Enter the WAN interface name or 'a' for auto-detection
(em0 em1 em2 em3 em1.10 or a): em0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(em1 em2 em3 em1.10 a or nothing if finished): em1

Enter the Optional 1 interface name or 'a' for auto-detection
(em2 em3 em1.10 a or nothing if finished): em1.10

Enter the Optional 2 interface name or 'a' for auto-detection
(em2 em3 a or nothing if finished):

The interfaces will be assigned as follows:

WAN  -> em0
LAN  -> em1
OPT1 -> em1.10

Do you want to proceed [y|n]? y
```

10.
11.

```
Starting syslog...done.
Starting CRON... done.
pfSense 2.8.1-RELEASE amd64 20251215-1731
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: 353e1d94a6db92b5b901

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

 WAN (wan) -> em0 -> v4/DHCP4: 10.253.151.59/24
 LAN (lan) -> em1 -> v4: 192.168.10.254/24

 0) Logout / Disconnect SSH          9) pfTop
 1) Assign Interfaces               10) Filter Logs
 2) Set interface(s) IP address     11) Restart GUI
 3) Reset admin account and password 12) PHP shell + pfSense tools
 4) Reset to factory defaults       13) Update from console
 5) Reboot system                   14) Enable Secure Shell (sshd)
 6) Halt system                     15) Restore recent configuration
 7) Ping host                       16) Restart PHP-FPM
 8) Shell

Enter an option:
```

```
em0     08:00:27:3a:35:9d   (up) Intel(R) Legacy PRO/1000 MT 82540EM
em1     08:00:27:c9:8f:ea   (up) Intel(R) Legacy PRO/1000 MT 82540EM
em2     08:00:27:e6:1c:76 (down) Intel(R) Legacy PRO/1000 MT 82540EM
em3     08:00:27:95:ec:74 (down) Intel(R) Legacy PRO/1000 MT 82540EM

Do VLANs need to be set up first?
If VLANs will not be used, or only for optional interfaces, it is typical to
say no here and use the webConfigurator to configure VLANs later, if required.

Should VLANs be set up now [y|n]? y

WARNING: all existing VLANs will be cleared if you proceed!

Do you want to proceed [y|n]? y



VLAN Capable interfaces:

em0     08:00:27:3a:35:9d   (up)
em1     08:00:27:c9:8f:ea   (up)
em2     08:00:27:e6:1c:76
em3     08:00:27:95:ec:74

Enter the parent interface name for the new VLAN (or nothing if finished): em1
```

```
em1     08:00:27:c9:8f:ea   (up) Intel(R) Legacy PRO/1000 MT 82540EM
em2     08:00:27:e6:1c:76 (down) Intel(R) Legacy PRO/1000 MT 82540EM
em3     08:00:27:95:ec:74 (down) Intel(R) Legacy PRO/1000 MT 82540EM

Do VLANs need to be set up first?
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WARNING: all existing VLANs will be cleared if you proceed!

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em0     08:00:27:3a:35:9d   (up)
em1     08:00:27:c9:8f:ea   (up)
em2     08:00:27:e6:1c:76
em3     08:00:27:95:ec:74

Enter the parent interface name for the new VLAN (or nothing if finished): em1
Enter the VLAN tag (1-4094): 10
```

```
VLAN Capable interfaces:

em0     08:00:27:3a:35:9d   (up)
em1     08:00:27:c9:8f:ea   (up)
em2     08:00:27:e6:1c:76
em3     08:00:27:95:ec:74

Enter the parent interface name for the new VLAN (or nothing if finished):


VLAN interfaces:

em1.10          VLAN tag 10, parent interface em1

If the names of the interfaces are not known, auto-detection can
be used instead. To use auto-detection, please disconnect all
interfaces before pressing 'a' to begin the process.

Enter the WAN interface name or 'a' for auto-detection
(em0 em1 em2 em3 em1.10 or a): em0

Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(em1 em2 em3 em1.10 a or nothing if finished): em1
```

```
em0     08:00:27:3a:35:9d   (up)
em1     08:00:27:c9:8f:ea   (up)
em2     08:00:27:e6:1c:76
em3     08:00:27:95:ec:74

Enter the parent interface name for the new VLAN (or nothing if finished):


VLAN interfaces:

em1.10          VLAN tag 10, parent interface em1

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Enter the LAN interface name or 'a' for auto-detection
NOTE: this enables full Firewalling/NAT mode.
(em1 em2 em3 em1.10 a or nothing if finished): em1

Enter the Optional 1 interface name or 'a' for auto-detection
(em2 em3 em1.10 a or nothing if finished): em1.10
```
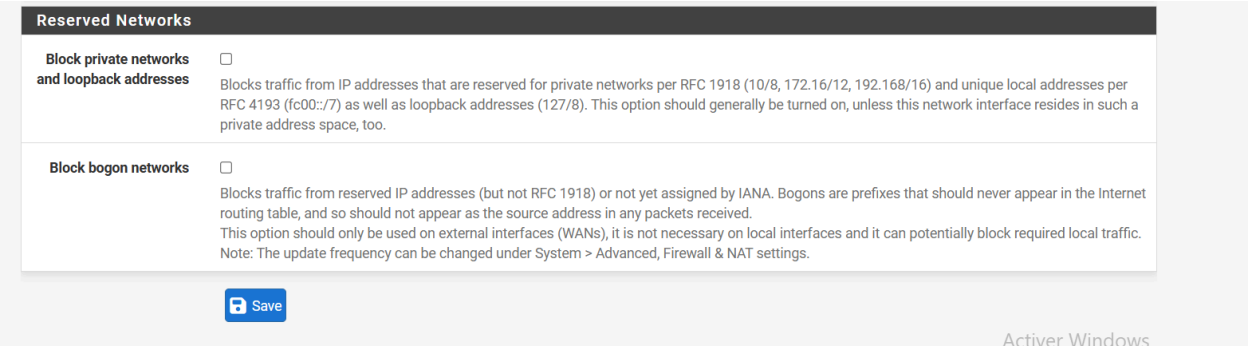
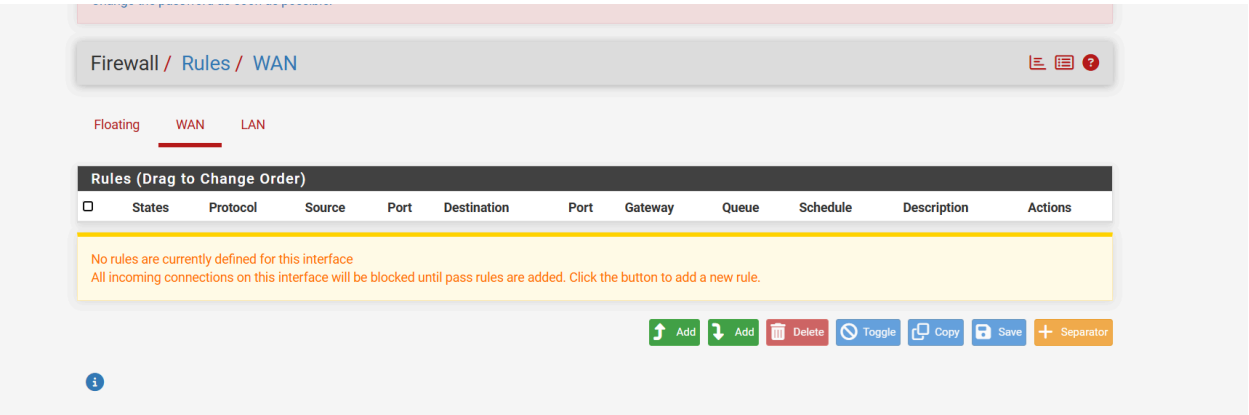

Maintenant nous allons passer par le protocole HTTPS. Dans Système → advanced → Accès administrateur. Cocher la case HTTPS.
TCP port on change le numéro de port de 443 celui de https à 20031 customized port → save

Etape 2 ouvrir le port de pfsnes pour qu' on puisse avoir a accès à distance

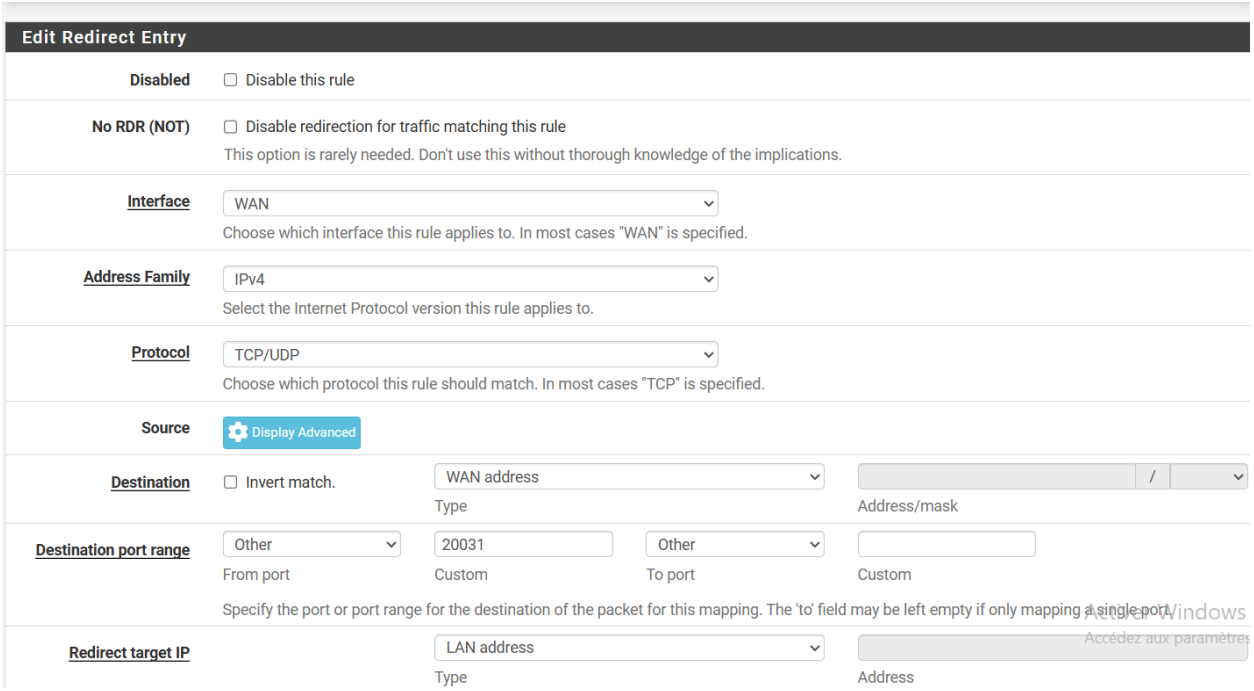
1. Interface → Wan → Block private networks and loopback addresses—> dechocher → Block bogon networks → decocher → save → apply changes



2.
3. Firewall → NAT → Port Forward → Clique + Add



4.
5. Interface : WAN
6. Protocol : TCP
7. Destination : WAN address
8. Destination port range : From 20031 To 20031
9. Redirect target IP : Adresse
10. Description : ouvrir pfsens port
11. Save → apply changes

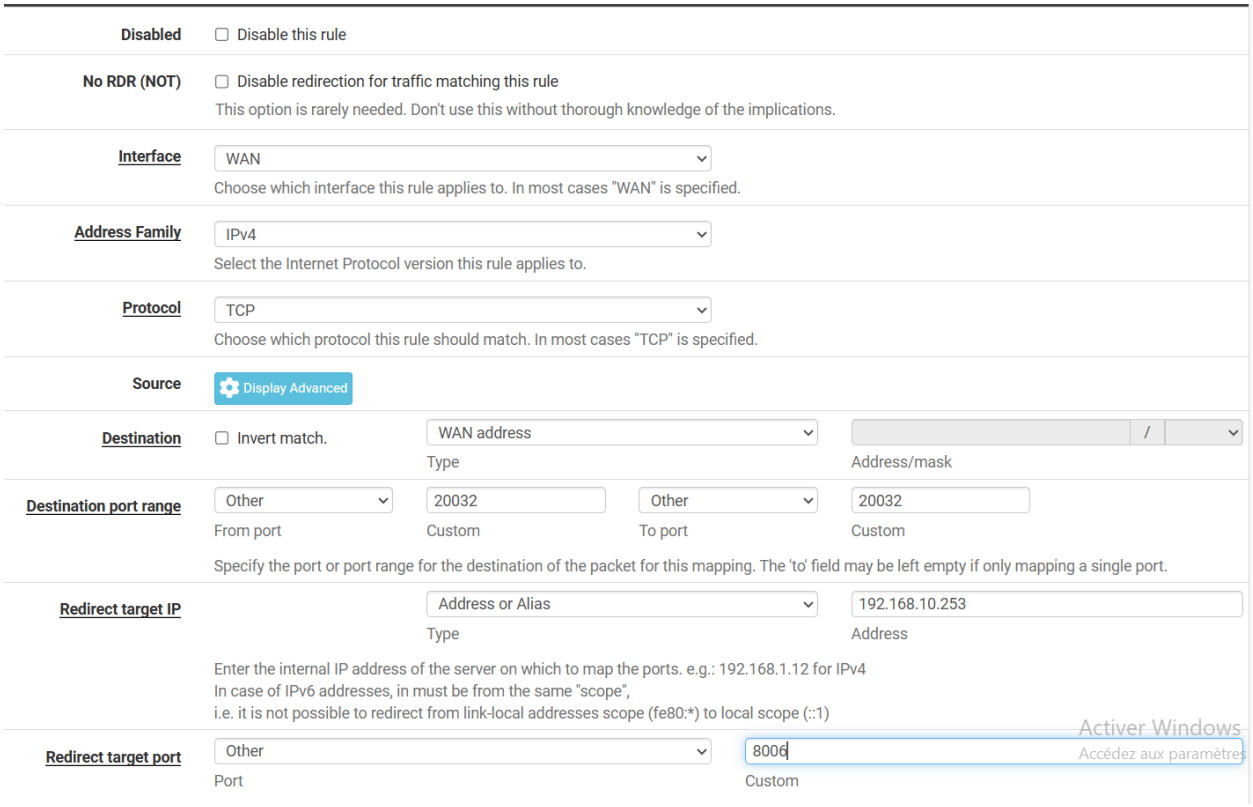
12. 

13. Va dans Firewall → Rules → WAN

- La règle NAT crée automatiquement la règle firewall correspondante (en bas de liste)
- Vérifie qu'elle existe (sinon ajoute-la manuellement)

Etape 2 ouvrir port proxmox

1. Firewall → NAT → Port Forward → + Add
2. Interface : WAN
3. Protocol : TCP
4. Destination : WAN address
5. Destination port range : From 8007 To 8007 (choisis un port différent de 8006 !)
6. Redirect target IP : 192.168.1.243 ← IP de Proxmox host
7. Redirect target port : 8006
8. Description : ouvrir Proxmox port
9. Save → apply changes



10.

11. Va dans Firewall → Rules → WAN

- La règle NAT crée automatiquement la règle firewall correspondante (en bas de liste)
- Vérifie qu’elle existe (sinon ajoute-la manuellement)